Individual Coursework Submission Form

Specialist Masters Programme

Surname: PINTO	**First Name: AARON NEIL**
MSc in: ACTUARIAL SCIENCE	**Student ID number: 240053223**
Module Code: SMM065	
Module Title: CONTINGENCIES (SUBJECT CM1(2))	
Lecturer: DOUGLAS WRIGHT	**Submission Date: 13 MARCH 2025**

Declaration:
By submitting this work, I declare that this work is entirely my own except those parts duly identified and referenced in my submission. It complies with any specified word limits and the requirements and regulations detailed in the coursework instructions and any other relevant programme and module documentation. In submitting this work, I acknowledge that I have read and understood the regulations and code regarding academic misconduct, including that relating to plagiarism, as specified in the Programme Handbook. I also acknowledge that this work will be subject to a variety of checks for academic misconduct.

We acknowledge that work submitted late without a granted extension will be subject to penalties, as outlined in the Programme Handbook. Penalties will be applied for a maximum of five days lateness, after which a mark of zero will be awarded.

Marker's Comments (if not being marked on-line):

Deduction for Late Submission:

Final Mark: %

MSc Actuarial Science 2024-25
Computer-Based Assessment
SMM065 contingencies cm1(2)

Aaron Neil Pinto

13 March 2025

Contents

1 Mortality Table Extensions **4**
- 1.1 Methodology for Extending the Mortality Table . . . 4
- 1.2 Verification of ${}_5p_{[60]}$ and ${}_5p_{60}$ . . . 4
- 1.3 Calculation of Curtate Expectation of Life e_x . . . 5
- 1.4 Extending the curatate expected life tables $e_{[x]}$ and $e_{[x-1]+1}$ . . . 6
- 1.5 Adjusting Mortality Rates by 10% . . . 6
- 1.6 Change in e_{60} After Mortality Improvements . . . 7

2 Question 2: Commutation Functions **9**
- 2.1 Calculating Commutation Functions D_x and C_x . . . 9
- 2.2 Calculating Further Commutation Functions N_x, S_x, M_x, and R_x . . . 10

3 Question 3: Insurance Pricing with Commutation Functions **12**
- 3.1 Whole of Life Assurance Premium . . . 12
- 3.2 15-Year Term and Endowment Assurance Premium . . . 12
- 3.3 Premium Comparison . . . 14

4 Question 4: Adjusting for Higher Interest Rates **15**
- 4.1 Recalculating Premiums After Interest Rate Increase . . . 15
- 4.2 Explanation of Premium Reduction Due to Higher Interest Rates . . . 16
- 4.3 Percentage Reduction in Premiums . . . 16
- 4.4 Relative Change in Premiums for Different Contracts . . . 17

5 Question 5: Prospective Reserves Calculation **18**
- 5.1 Whole of Life Assurance Reserve at Duration t . . . 18
- 5.2 Term Assurance Reserve at Duration t . . . 19
- 5.3 Endowment Assurance Reserve at Duration t . . . 20
- 5.4 Graph of Prospective Reserves . . . 21

6 Question 6: Recalculating Prospective Reserves After Interest Rate Adjustment **22**
- 6.1 Prospective Reserves with Revised Premiums . . . 22
- 6.2 Increased Initial Reserve at t=0 and Its Impact on the Insurance Company's Operations . . . 23

A Appendix **24**

1 Mortality Table Extensions

1.1 Methodology for Extending the Mortality Table

We construct the life tables using the given graduated probability q_x values from the AM92 tables to calculate the corresponding l_x values for ages 17 to 120. We begin by selecting an arbitrary radix, which represents the number of individuals alive at age 17 (the starting age). We set $l_{17} = 10,000$. The number of survivors at each age can be calculated using the relation:

$$A^{1}_{x:\overline{n}|}$$

$$p_x = \frac{l_{x+1}}{l_x}$$

Since $p_x + q_x = 1$, we can express p_x as:

$$p_x = 1 - q_x$$

Using this expression, the number of survivors at age $x + 1$ is determined by the following equation:

$$\boldsymbol{l_{x+1} = l_x \cdot p_x = l_x \cdot (1 - q_x)} \tag{1}$$

We apply this recursive formula iteratively to compute l_x for each subsequent age. The process continues until we reach the desired age of 120.

Example:

For $\mathbf{x = 19}$: from equation (1), we have:

$$l_{19} = l_{18} \cdot (1 - q_{18})$$

Previously determined $l_{18} = 9994$ and $q_{18} = 0.000594$, we obtain:

$$l_{19} = 9994 \cdot (1 - 0.000594) = \mathbf{9988.0636}$$

This recursive procedure is repeated for each age until the table reaches the target age of 120. The preview of the Excel computation is shown in Appendix.

1.2 Verification of ${}_5p_{[60]}$ and ${}_5p_{60}$

To verify the answers, we use the following Table 1 :

Age x	$l_{[x]}$	l_x
60	9263.1422	9287.2164
⋮	⋮	⋮
65	8772.7359	8821.2612

Table 1: Values of $l_{[x]}$ and l_x for each age x

Calculate ${}_5p_{[60]}$: The probability that a life selected at age 60 survives for 5 more years is given by:

$$ {}_5p_{[60]} = \frac{l_{65}}{l_{[60]}} $$

Using the values from table (1), we have: Substituting these values:

$$ {}_5p_{[60]} = \frac{8821.2612}{9263.1422} = \mathbf{0.952296853} $$

Calculate ${}_5p_{60}$: The probability that a person aged 60 survives for five more year is given by:

$$ {}_5p_{60} = \frac{l_{65}}{l_{60}} $$

Using the values from part (i), suppose we have:

$$ l_{65} = 8821.2612 \quad \text{and} \quad l_{60} = 9287.2164 $$

Substituting these values:

$$ {}_5p_{60} = \frac{8821.2612}{9287.2164} = \mathbf{0.949828323} $$

1.3 Calculation of Curtate Expectation of Life e_x

The curtate future lifetime K_x represents the number of complete years lived after exact age x. The expected curtate future lifetime, or curtate expectation of life, is given by:

$$ e_x = E(K_x) = \sum_{k=1}^{\infty} k \; p_x = \frac{l_{x+1} + l_{x+2} + \dots}{l_x} $$

where l_x represents the number of survivors at age x from a AM92 life table.

Calculation of e_{17}:
For $x = 17$, using the AM92 life table:

$$ e_{17} = \frac{l_{18} + l_{19} + l_{20} + \dots}{l_{17}} $$

From Figure 1, we obtain the values:

$$ e_{17} = \frac{(9994.00 + 9988.0636 + \dots)}{10000.0000} $$

Thus, we get:

$$ e_{17} = 61.339 $$

This process is repeated until we reach the required age of 120. The Excel sheet preview is shown in Appendix.

1.4 Extending the curatate expected life tables $e_{[x]}$ and $e_{[x-1]+1}$

we use the similar approach for the as we used in section 1.3. For select mortality, we adjust the expectation formula using the select life table:

$$e_{[x-1]+1} = \sum_{k=1}^{\infty} k\, p_{[x-1]+1} = \frac{l_{x+1} + l_{x+2} + \dots}{l_{[x-1]+1}} \tag{2}$$

$$e_{[x]} = E(K_{[x]}) = \sum_{k=1}^{\infty} k\, p_{[x]} = \frac{l_{[x]+1} + l_{x+2} + \dots}{l_{[x]}} \tag{3}$$

The AM92 table has a 2-year select period, meaning that the selection effect only applies for the first two years. After this period,the selection effect no longer applies as we observe in eqution (2) and (3).

$$l_{[x]} \to l_{[x]+1} \to l_{x+2} \to l_{x+3} \to \dots$$

Calculation of $e_{[x-1]+1}$ for $x = 20$:

To calculate $e_{[x-1]+1}$ for a specific age $x = 20$, we use the following formula based on the select life table:

$$e_{[x-1]+1} = e_{[19]+1} = \frac{l_{21} + l_{22} + l_{23} + \dots}{l_{[20-1]+1}}$$

$$e_{[19]+1} = \frac{(9976.3909 + 9970.6346 + \dots)}{9976.0016} = 58.449$$

Calculation of $e_{[x]}$ for $x = 20$:

Here,the select effect has already been removed after 2 years, we proceed with the formula for the ultimate life table:

$$e_{[20]} = \frac{l_{[21]} + l_{22} + l_{23} + \dots}{l_{[20]}}$$

$$e_{[20]} = \frac{99974.5046 + 9970.6346 + \dots}{9980.2432} = 58.458$$

The Excel sheet preview is shown in Appendix.

1.5 Adjusting Mortality Rates by 10%

$$\boldsymbol{q_x^{new} = q_x \cdot (1 - 0.10)} \tag{4}$$

This process is repeated for each year, unntil we reach the required age of 120. Then we extend these values to construct life table and expected curtate table as shown previous sections. The Excel sheet preview is shown in Appendix.

Example:
For $\mathbf{x = 17}$: from equation (4), we have:

$$\boldsymbol{q_x^{new} = \cdot(1 - 0.10)} = 0.00054$$

Now, we use the new tables to calculate the following:
Calculate ${}_5p_{[60]}$: The probability that a life selected at age 60 survives for 5 more years is given by:

$$ {}_5p_{[60]} = \frac{l_{65}}{l_{[60]}} $$

Substituting these values:

$$ {}_5p_{[60]} = \frac{8932.9162}{9334.43885} = \mathbf{0.956984815} $$

Calculate ${}_5p_{60}$: The probability that a person aged 60 survives for five more year is given by:

$$ {}_5p_{60} = \frac{l_{65}}{l_{60}} $$

Substituting these values:

$$ {}_5p_{60} = \frac{9356.2538}{8932.9162} = \mathbf{0.954753516} $$

	Original	Reduced 10%
${}_5p_{[60]}$	0.952296853	0.956984815
${}_5p_{60}$	0.949828323	0.954753516

Table 2: Comparing original and after reduced 10%

From Table (2) we can clearly see that both survival probabilities (ultimate and select)increased after the 10% mortality reduction.This means that more individuals will reach older ages than previously expected. The increase in ${}_5p_{[60]}$ from 0.9523 to 0.9570 and $5p_{60}$ from 0.9498 to 0.9548 suggests that the likelihood of surviving the next 5 years has improved. The difference is small but significant—even a minor change in survival rates can have substantial effects on pension schemes and life insurance pricing.In actuarial terms, this could lead to:Longer pension payouts for retirees, Higher annuity liabilities for insurance companies. Changes in premium pricing for life insurance and pension funds.

1.6 Change in e_{60} After Mortality Improvements

Using the same formulas presented in Sections 1.3 and 1.4 and extending the table to account for a 10% reduction in mortality, we obtain the following results for life aged 60:

Scenario	e_{60} (Curtate Expectation of Life)
Original Value	20.670
After Reduction	21.559

Table 3: Comparison of the Curtate Expectation of Life (e_{60}) Before and After the Suggestion

The impact of the mortality reduction extends beyond survival probabilities and also affects the curtate expectation of life, e_{60}. The revised calculations indicate that the expected curtate life currently aged 60 will live has increased from **20.670 to 21.559**. This increase of approximately **0.889 years** suggests that, due to lower mortality rates, individuals are expected to live longer on average.

For pension providers, a longer life expectancy means an extended duration of pension payouts, increasing the financial burden on retirement funds. Similarly, life insurance companies that offer annuities will need to reassess their pricing models, as policyholders will now receive payments for a longer period.

2 Question 2: Commutation Functions

2.1 Calculating Commutation Functions D_x and C_x

Using the standard actuarial formulas and assuming an interest rate of 4%, the commutation functions D_x and C_x are calculated as follows:

$$D_x = l_x \cdot v^x \tag{5}$$

$$C_x = v^{x+1} \left(l_x - l_{x+1} \right) \tag{6}$$

where l_x represents the number of lives at age x from the life table, and the discount factor is given by:

$$v = \frac{1}{1+i}, \quad \text{with } i = 0.04$$

This process is repeated for each year, using the life table data until we reach the required age of 110. By substituting the values of l_x and applying the formula for each age, the commutation functions can be computed for the entire range of ages.

Example for Both Select and Ultimate Calculations:
Calculation for D_x:

$$D_{20} = l_{20} \cdot v^{20} = 9982.2006 \cdot \frac{1}{1.04^{20}} = 4555.75$$

$$D_{[20-1]+1} = l_{[20-1]+1} \cdot v^{20} = 9976.0016 \cdot \frac{1}{1.04^{20}} = 4377.80$$

$$D_{[20]} = l_{[20]} \cdot v^{20} = 9980.2432 \cdot \frac{1}{1.04^{20}} = 4554.85$$

Calculations for C_x:
We have introduced an additional table to simplify the calculation i.e d_x. This table calculates the difference $l_x - l_{x+1}$(Refer the appendix for the Excel preview for d_x table). we have,

$$C_{20} = d_{20} \cdot v^{20+1} = 5.8096 \cdot \frac{1}{1.04^{21}} = 2.55$$

$$C_{[20-1]+1} = d_{[20-1]+1} \cdot v^{20+1} = 5.3671 \cdot \frac{1}{1.04^{21}} = 2.26$$

$$C_{[20]} = d_{[20]} \cdot v^{20+1} = 4.2416 \cdot \frac{1}{1.04^{21}} = 1.86$$

The Excel preview for the above computation is shown in appendix.

2.2 Calculating Further Commutation Functions N_x, S_x, M_x, and R_x

In addition to D_x and C_x, the following commutation functions are calculated as part of the extended process:

$$N_x = \sum_{y=x}^{\infty} D_y \tag{7}$$

$$S_x = \sum_{y=x}^{\infty} N_y \tag{8}$$

$$M_x = \sum_{y=x}^{\infty} C_y \tag{9}$$

$$R_x = \sum_{y=x}^{\infty} M_y \tag{10}$$

By using the above formulas, we can extend the calculations of the commutation functions over the age range and select mortality.

Calculation for N_x: Using the equation(7), we have:

1. For duration 2+:

$$N_{17} = D_{17} + D_{18} + \cdots + D_{110} = 5133.73 + 4933.32 + 4740.76 + \cdots = 119959.94$$

2. For duration 1:

$$N_{[17]+1} = D_{[17]+1} + D_{19} + \cdots = 4933.09 + 4740.76 + 4555.75 + \cdots = 114825.98$$

3. For duration 0:

$$N_{[17]} = D_{[17]} + D_{[17]+1} + D_{19} + \cdots = 5132.61 + 4933.09 + \cdots = 119958.58$$

The Excel preview for the above computation is shown in appendix.

Calculation for S_x: We use the equation (8), we have:

1. For duration 2+:

$$S_{17} = N_{17} + N_{18} + \cdots + = 2398087.20$$

2. For duration 1:

$$S_{[17]+1} = N_{[17]+1} + N_{19} + \cdots = 2278127.03$$

3. For duration 0:

$$S_{[17]} = N_{[17]} + N_{[17]+1} + N_{19} + \cdots = 2398085.62$$

The Excel preview for the above computation is shown in appendix.

Calculation for M_x: We use the equation (9), we have:

1. For duration 2+:

$$M_{17} = C_{17} + C_{18} + \cdots + = 519.89$$

2. For duration 1:

$$M_{[17]+1} = C_{[17]+1} + C_{19} + \cdots = 516.71$$

3. For duration 0:

$$M_{[17]} = C_{[17]} + C_{[17]+1} + C_{19} + \cdots = 518.82$$

The Excel preview for the above computation is shown in appendix.

Calculation for R_x: We use the equation (10), we have:

1. For duration 2+:

$$R_{17} = M_{17} + M_{18} + \cdots + = 27725.81$$

2. For duration 1:

$$R_{[17]+1} = M_{[17]+1} + M_{19} + \cdots = 27205.71$$

3. For duration 0:

$$R_{[17]} = M_{[17]} + M_{[17]+1} + M_{19} + \cdots = 27724.52$$

The Excel preview for the above computation is shown in appendix.

3 Question 3: Insurance Pricing with Commutation Functions

In Excel, the interest rate has been set at 3%, with the interest rate i defined as a variable using a data validation (list). This method eliminates the need for constructing a new set of tables, instead utilizing the existing commutation functions while allowing flexibility in adjusting the interest rate assumption.

3.1 Whole of Life Assurance Premium

The level annual premium for a whole of life assurance policy with a sum assured of £250,000, payable at the end of the year of death, is calculated using the formula:

$$P = \frac{S \cdot A_x}{\ddot{a}_x} \tag{11}$$

where S is the sum assured, A_x is the expected present value of a whole of life assurance and $\ddot{a}_x$ is the expected present value of a life annuity due for a life aged x.
The present value A_x and $\ddot{a}_x$using computation is given by:

$$A_x = \frac{M_x}{D_x}$$

$$\ddot{a}_x = \frac{N_x}{D_x}$$

Here, D_x, M_x, and N_x are commutation functions based on the AM92 ultimate mortality table.

For AM92 table with interest 3%, we have:

$$M_{50} = 942.67, \quad D_{50} = 2215.39, \quad N_{50} = 43696.71$$

Substituting these values into the equations:

$$A_{50} = \frac{942.67}{2215.39} = 0.42550973$$

$$\ddot{a}_{50} = \frac{43696.71}{2215.39} = 19.7241614$$

Now, using the formula (11) for the premium: We have S=250,000

$$P = \frac{S \cdot A_{50}}{\ddot{a}_{50}} = \frac{250,000 \cdot 0.42550973}{19.7241614} = 5393.26$$

Thus, the level annual premium P is **£5393.26**.

3.2 15-Year Term and Endowment Assurance Premium

The level annual premium for a 15-year term and endowment assurance policy with a sum assured of £250,000, payable at the end of the year of death if death occurs within

15 years or at the end of 15 years if the policyholder survives, is calculated using the general formula:

$$P^1_{x:\overline{n}|} = S \cdot \frac{A^1_{x:\overline{n}|}}{\ddot{a}_{x:\overline{n}|}} \tag{12}$$

$$P_{x:\overline{n}|} = S \cdot \frac{A_{x:\overline{n}|}}{\ddot{a}_{x:\overline{n}|}} \tag{13}$$

where S is the sum assured, $A_{x:\overline{n}|}$ is the expected present value of a n-year endowment assurance, $A^1_{x:\overline{n}|}$ is the expected present value of a n-year term assurance,and $\ddot{a}_{x:n}$ is the expected present value of a n-year temporary life annuity due. These can be calulated using computation function, using formula:

$$A_{x:\overline{n}|} = \frac{(M_x - M_{x+n}) + D_{x+n}}{D_x}$$

$$A^1_{x:\overline{n}|} = \frac{M_x - M_{x+n}}{D_x}$$

$$\ddot{a}_{x:\overline{n}|} = \frac{N_x - N_{x+n}}{D_x}$$

Using the AM92 Table with 3%:

$$M_{50} = 942.67 \quad M_{65} = 790.21 \quad D_{50} = 2215.39 \quad D_{65} = 1291.55 \quad N_{50} = 43696.71$$

$$N_{65} = 17212.48$$

Using these values we can calulate the expected prevent values as follows*:

$$A^1_{50:\overline{15}|} = \frac{M_{50} - M_{65}}{D_{50}} = \frac{942.67 - 790.21}{2215.39} = 0.06882$$

$$A_{50:\overline{15}|} = \frac{(M_{50} - M_{65}) + D_{65}}{D_{50}} = \frac{(942.67 - 790.21) + 1291.55}{2215.39} = 0.65181$$

$$\ddot{a}_{50:\overline{15}|} = \frac{N_{50} - N_{65}}{D_{50}} = \frac{43696.71 - 17212.48}{2215.39} = 11.955$$

Now, we calulate the premuim for term assurance:

$$P^1_{50:\overline{15}|} = S \cdot \frac{A^1_{50:\overline{15}|}}{\ddot{a}_{50:\overline{15}|}} = 250,000 \cdot \frac{0.06882}{11.955} = 1439.1468$$

Now, we calulate the premuim for term assurance:

$$P_{50:\overline{15}|} = S \cdot \frac{A_{50:\overline{15}|}}{\ddot{a}_{50:\overline{15}|}} = 250,000 \cdot \frac{0.65181}{11.955} = 13630.82$$

The annual premium for the 15-year term assurance is **£1,439.14**, and the annual premium for the 15-year endowment assurance is **£13,630.82**.

[0]For the purpose of simplifying the calculations, a table has been created in Excel to compute the present values of the relevant functions. This table incorporates a variable interest rate, which can be adjusted as necessary.

3.3 Premium Comparison

The premiums for three types of policies are influenced by the duration and certainty of the liability. The **term assurance** has the lowest premium of **£1,439.14** because it only covers a limited period of 15 years, with no payout if the policyholder survives. The **whole of life assurance** premium is **£5,393.26**, as it offers lifetime coverage and a guaranteed payout upon death, making it more costly. The **endowment assurance** combines both term assurance and pure endowment benefits (a payout if the policyholder survives) with premuim **£13630.82**. This dual benefit results in a higher premium since it guarantees a payout either upon death or at the end of the policy term.

In conclusion, the premiums vary based on the duration and likelihood of payout. The **term assurance** is the least expensive, the **whole of life assurance** falls in the middle, and the **endowment assurance** is the most costly due to its guaranteed payout.

4 Question 4: Adjusting for Higher Interest Rates

4.1 Recalculating Premiums After Interest Rate Increase

To calculate the revised level annual premiums, we need to adjust the present values for each of the three contracts using the new interest rate of 4.75%. We will use formulas (11), (12), and (13) to calculate the revised premiums.

Using the AM92 Table with 3%:

$$M_{50} = 261.29 \quad M_{65} = 205.17 \quad D_{50} = 954.14 \quad D_{65} = 432.03 \quad N_{50} = 15279.17$$

$$N_{65} = 5003.00$$

Using these values we can calulate the expected prevent values as follows:

$$A_{50} = \frac{M_{50}}{D_{50}} = \frac{942.67}{954.14} = 0.9879787033$$

$$\ddot{a}_{50} = \frac{N_{50}}{D_{50}} = \frac{15279.17}{954.14} = 16.01355147$$

$$A^{1}_{50:\overline{15}|} = \frac{M_{50} - M_{65}}{D_{50}} = \frac{261.29 - 205.17}{954.14} = 0.05882$$

$$A_{50:\overline{15}|} = \frac{(M_{50} - M_{65}) + D_{65}}{D_{50}} = \frac{(261.29 - 205.17) + 432.03}{954.14} = 0.51162$$

$$\ddot{a}_{50:\overline{15}|} = \frac{N_{50} - N_{65}}{D_{50}} = \frac{15279.17 - 5003.00}{954.14} = 10.770$$

Now, we calulate the premuim for term assurance:

$$P = \frac{S \cdot A_{50}}{\ddot{a}_{50}} = \frac{250,000 \cdot 0.9879787033}{16.01355147} = 4275.19$$

Now, we calulate the premuim for term assurance:

$$P^{1}_{50:\overline{15}|} = S \cdot \frac{A^{1}_{50:\overline{15}|}}{\ddot{a}_{50:\overline{15}|}} = 250,000 \cdot \frac{0.05882}{10.770} = 1365.24$$

Now, we calulate the premuim for term assurance:

$$P_{50:\overline{15}|} = S \cdot \frac{A_{50:\overline{15}|}}{\ddot{a}_{50:\overline{15}|}} = 250,000 \cdot \frac{0.51162}{10.770} = 11875.83$$

The annual level premium for the whole life assurance is **£4,275.19**. For the 15-year term assurance, the annual premium amounts to **£1,365.24**, while for the 15-year endowment assurance, it is **£11,875.83**.

4.2 Explanation of Premium Reduction Due to Higher Interest Rates

An increase in the pricing basis interest rate from 3.0% per annum to 4.75% per annum leads to a higher discounting effect on future benefit payouts. This reduces their present value, thereby lowering the amount of premium required to fund these benefits.

$$v = (1+i)^{-1}$$

As the interest rate i increases, the discount factor v decreases because the denominator $(1+i)$ becomes larger, reducing the overall value of v.

Since premiums are designed to accumulate with interest over time to meet future liabilities, a higher interest rate means that a smaller premium is needed to achieve the same financial outcome. As a result, all three contracts—whole life assurance, term assurance, and endowment assurance—have lower premiums than at 3

4.3 Percentage Reduction in Premiums

For each of the three contracts, the percentage reduction in the premium can be calculated using the following formula:

$$\text{Percentage Reduction} = \frac{\text{Old Premium} - \text{New Premium}}{\text{Old Premium}} \times 100 \tag{14}$$

Using the equation (13): We have,

1. Whole Life Assurance:

$$\text{Percentage Reduction} = \frac{5,393.26 - 4,275.19}{5,393.26} \times 100 = \frac{1,118.07}{5,393.26} \times 100 \approx 20.74\%$$

2. 15-Year Term Assurance:

$$\text{Percentage Reduction} = \frac{1,439.14 - 1,365.24}{1,439.14} \times 100 = \frac{73.90}{1,439.14} \times 100 \approx 5.14\%$$

3. 15-Year Endowment Assurance:

$$\text{Percentage Reduction} = \frac{13,630.82 - 11,875.83}{13,630.82} \times 100 = \frac{1,755.00}{13,630.82} \times 100 \approx 12.88\%$$

Contract Type	**Premium at 3.0% (£)**	**Premium at 4.75% (£)**	**Percentage Reduction (%)**
Whole Life Assurance	**5,393.26**	**4,275.19**	**20.74%**
15-Year Term Assurance	**1,439.14**	**1,365.24**	**5.14%**
15-Year Endowment Assurance	**13,630.82**	**11,875.83**	**12.88%**

Table 4: Comparison of Annual Premiums at 3.0% and 4.75% with Percentage Reduction

4.4 Relative Change in Premiums for Different Contracts

The interest rates play a cruital role in determining the premiums when pricing life insurance products. An increase in the rate of interest leads to a reduction in the premiums across all types of life contracts. However, the magnitude of this reduction varies depending on the nature and the duration of the contract.

The fundamental reason for this decrease in premuims is the effect of discounting. Higher the rate of interest results into lower expected present value of the future benefits payouts .Since premiums are set to cover these payouts, a reduction in the present value of benefits means lower required premiums. However, the extent to which premiums decrease depends on how far into the future benefits are expected to be paid.

For whole life assurance, the benefits are typically paid many years into the future. Since these distant payouts are heavily affected by discounting, a small change in the interest rate can lead to a significant reduction in the present value of the benefits. As a result, the required premiums for whole life assurance decreases as interest rate increases.

Term assurance provides coverage only for a fixed period. Since the benefits, if paid, are expected to occur relatively soon, they are less affected by discounting. Due to this, the reduction in the present value of benefits—thus the decrease in premiums—is smaller compared to whole life assurance. In contrast, the longer the expected duration before a benefit is paid, the more sensitive its present value is to changes in the discount rate.

5 Question 5: Prospective Reserves Calculation

5.1 Whole of Life Assurance Reserve at Duration t

For a whole life assurance contract issued to a life of exact age 50, the prospective reserve at duration t, ${}_tV_x^{(WL)}$, is given by the following equation:

$$ {}_tV_x^{(WL)} = S \cdot A_{x+t} - P_{WL} \cdot \ddot{a}_{x+t} \tag{15} $$

Where:

- S is the benefit amount payable upon death.
- $v = \frac{1}{1+i}$ is the discount factor, where $i = 0.03$ is the interest rate.
- A_{x+t} represents the expected present value for a life aged $x + t$.
- $\ddot{a}_{x+t}$ is the present value of an whole life annuity.
- P is the annual premium.

Example:
For time 1: $S = 250,000$, $P = 5,393.26$, $A_{51} = 0.436862$, $\ddot{a}_{51} = 19.334$, $i = 0.03$
We have,

$$ {}_5V_{50}^{(WL)} = 250,000 \cdot 0.436862 - 5,393.26 \cdot 19.334 = 109215.60 - 104275.16 = 4940.45 $$

t	EPV benefit	EPV premium	${}_tV_{50}^{(\mathrm{WLA})}$
0	106377.37	106377.37	0.00
1	109215.60	104275.16	4940.45
2	112104.72	102135.25	9969.47
3	115042.48	99959.33	15083.15
4	118026.70	97748.98	20277.71
5	121054.81	95506.13	25548.68
6	124123.92	93232.92	30891.00
7	127230.72	90931.78	36298.94
8	130371.74	88605.30	41766.44
9	133543.16	86256.30	47286.86
10	136740.78	83887.90	52852.89
11	139960.27	81503.30	58456.97
12	143196.89	79106.01	64090.88
13	146445.65	76699.72	69745.93
14	149701.23	74288.38	75412.85

Table 5: Whole of Life Assurance Reserve at Duration t

5.2 Term Assurance Reserve at Duration t

For a term assurance contract issued to a life of exact age 50, the prospective reserve at duration t, ${}_tV_x^{(TA)}$, is given by the following equation:

$$ {}_tV_x^{(TA)} = S \cdot A^{\ 1}_{x+t:\overline{n-t}|} - P_{TA} \cdot \ddot{a}_{x+t:\overline{n-t}|} \tag{16} $$

Where:

- $A^{\ 1}_{x+t:\overline{n-t}|}$ represents the present value of a benefit payable for term assurance.
- $\ddot{a}_{x+t:\overline{n-t}|}$ is the present value of an annuity for term assurance.

Example:
For time 1: $S = 250,000$, $P = 1439.14$, $A^{\ 1}_{51:\overline{14}|} = 0.06855$, $\ddot{a}_{51:\overline{14}|} = 11.312$
We have,

$$ {}_1V_{50}^{(TA)} = 250,000 \cdot 0.06855 - 1439.14 \cdot 11.312 = 17136.47 - 16279.01 = 857.46 $$

t	EPV benefit	EPV premium	${}_tV_{50}^{(\mathrm{TA})}$
0	17204.37	17204.37	0.00
1	17136.47	16279.01	857.46
2	16996.06	15328.13	1667.93
3	16770.80	14350.90	2419.91
4	16447.39	13346.35	3101.04
5	16010.46	12313.38	3697.08
6	15442.54	11250.75	4191.79
7	14723.55	10157.01	4566.55
8	13830.90	9030.43	4800.48
9	12738.75	7869.01	4869.74
10	11417.43	6670.40	4747.03
11	9833.34	5431.77	4401.56
12	7947.69	4149.80	3797.89
13	5715.92	2820.51	2895.41
14	3086.41	1439.14	1647.27

Table 6: Term Assurance Reserve at Duration t

5.3 Endowment Assurance Reserve at Duration t

For an endowment assurance contract issued to a life of exact age 50, the prospective reserve at duration t, ${}_tV_x^{(EA)}$, is given by the following equation:

$$ {}_tV_x^{(EA)} = S \cdot A_{x+t:\overline{n-t}|} - P_{EA} \cdot \ddot{a}_{x+t:\overline{n-t}|} \tag{17} $$

Where:

- $A_{x+t:\overline{n-t}|}$ represents the Expected present value for endowment assurance.
- $\ddot{a}_{x+t:\overline{n-t}|}$ is the Expected present value of an annuity for the duration of the contract.

Example:
For time 1: $S = 250,000$, $P = 13630.82$, $A_{51:\overline{14}|} = 0.67053$, $\ddot{a}_{51:\overline{14}|} = 11.312$
We have:

$$ {}_5V_{50}^{(EA)} = 250,000 \cdot 0.67053 - 13630.82 \cdot 11.312 = 167633.60 - 154187.13 = 13446.47 $$

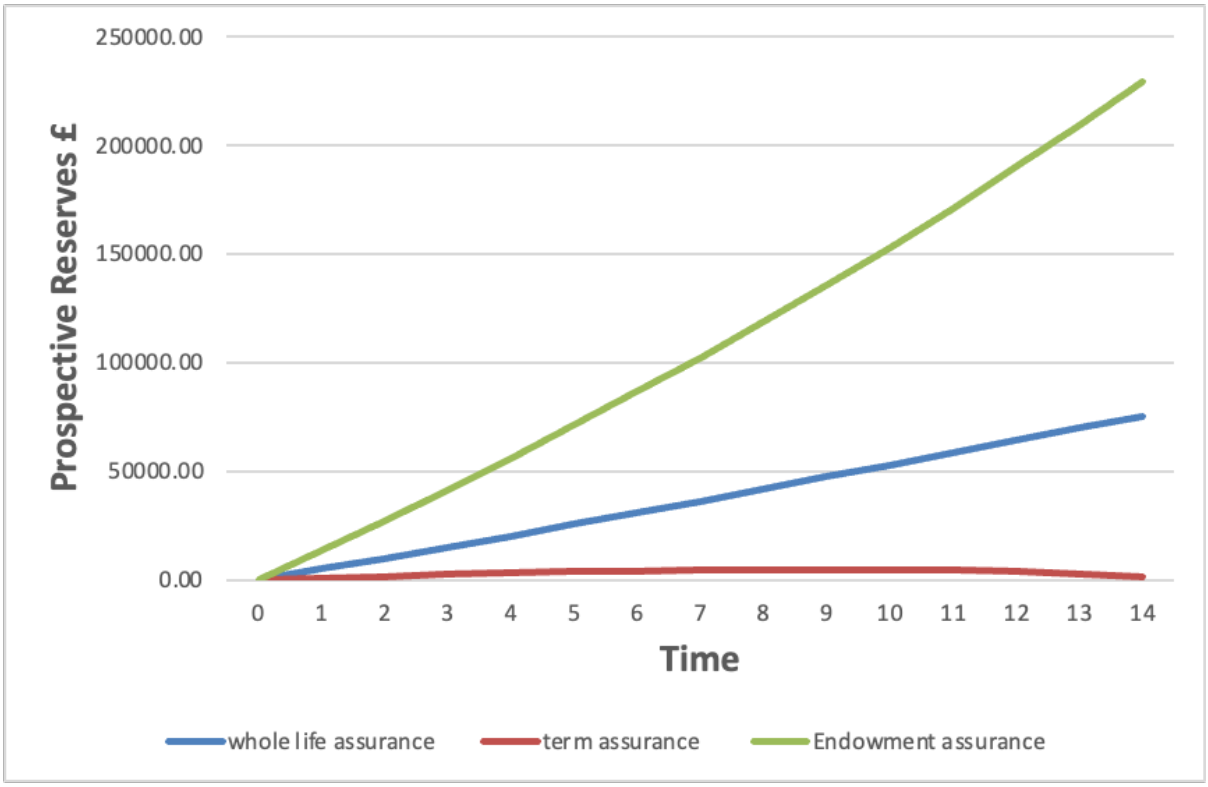
t	EPV benefit	EPV premium	${}_tV_{50}^{(\mathrm{EA})}$
0	162951.62	162951.62	0.00
1	167633.60	154187.13	13446.47
2	172444.75	145180.81	27263.95
3	177389.22	135924.92	41464.31
4	182471.92	126410.29	56061.63
5	187698.36	116626.55	71071.81
6	193074.92	106561.83	86513.09
7	198608.93	96202.35	102406.57
8	204309.04	85531.93	118777.10
9	210185.41	74531.57	135653.84
10	216249.99	63178.87	153071.13
11	222517.03	51447.20	171069.83
12	229003.38	39304.97	189698.41
13	235729.17	26714.51	209014.67
14	242718.45	13630.82	229087.62

Table 7: Assurance Reserve at Duration t

5.4 Graph of Prospective Reserves

The prospective reserves for three different insurance contracts—Whole Life Assurance, Term Assurance, and Endowment Assurance—have been analyzed over a 14-year period. The results show distinct trends in reserve accumulation across these contracts. The

Figure 1: Plot for prospective Reserves for Different Insurance Contracts



Endowment Assurance contract exhibits the steepest increase in reserves, accumulating the highest reserve levels over time. This is because Endowment Assurance guarantees a payout either at the end of the contract term or upon death, creating a need for faster and more substantial reserve accumulation to ensure that the policy is fully funded for the eventual payout.

The whole Life Assurance follows a moderate and steady reserve growth trajectory. Since this policy guarantees a death benefit at some uncertain point in the future, the reserves grow gradually over time to meet liabilities.

Meanwhile, Term Assurance maintains minimal reserves, with its reserve line remaining relatively flat throughout the period. This occurs because Term Assurance provides coverage only for a specified period, and many policies will likely expire without a claim. Therefore, insurers do not need to accumulate large reserves.

6 Question 6: Recalculating Prospective Reserves After Interest Rate Adjustment

6.1 Prospective Reserves with Revised Premiums

In this section, we recalculate the prospective reserves for three types of assurance contracts—Whole Life Assurance (WL), Term Assurance (TA), and Endowment Assurance (EA)—after adjusting for the change in interest rates. The revised premiums are based on the new interest rate of 4.75%, while the expected present value (EPV) calculations for each contract continue to use an interest rate of 3%, consistent with the valuation basis used in Question 5. The approach used in the recalculation is similar for all three contracts. The reserves are calculated using the following formulas:

For the whole life assurance contract:

$$ {}_tV_x^{(WL)} = S \cdot A_{x+t}^{3\%} - P_{WL}^{4.75\%} \cdot \ddot{a}_{x+t}^{3\%} $$

For the 15-year term assurance contract:

$$ {}_tV_{50}^{(TA)} = S \cdot A_{50+t:\overline{15-t|}}^{\;\;1\;\;\;

t	EPV benefit	EPV prem	tV50(EA)
0	162951.62	141971.32	20980.30
1	167633.60	134335.27	33298.33
2	172444.75	126488.53	45956.22
3	177389.22	118424.35	58964.87
4	182471.92	110134.74	72337.18
5	187698.36	101610.68	86087.68
6	193074.92	92841.81	100233.11
7	198608.93	83816.14	114792.79
8	204309.04	74519.55	129789.49
9	210185.41	64935.50	145249.91
10	216249.99	55044.48	161205.51
11	222517.03	44823.28	177693.74
12	229003.38	34244.38	194759.00
13	235729.17	23274.97	212454.21
14	242718.45	11875.83	230842.62

Table 8: Endowment Assurance

6.2 Increased Initial Reserve at t=0 and Its Impact on the Insurance Company's Operations

Following the increase in interest rates from 3.0% to 4.75%, the initial reserve at $t = 0$ for each insurance contract is now greater than zero due to the mismatch between the **valuation basis** (3%) and the **premium rate assumption** (4.75%).

For all three contracts, premiums are calculated using the 4.75% rate, which results in lower premiums compared to what would be required at the 3% rate. However, the future liabilities are still valued at the 3% rate, which leads to a higher present value of liabilities than the premiums collected. This mismatch results in a positive initial reserve being required to cover the difference between the lower premiums and the higher future liabilities.

The initial reserve was previously zero because, under the original assumptions (interest rate of 3.0%), the premiums collected for each contract were exactly sufficient to meet the present value of future liabilities. In other words, the present value of future benefits, calculated at 3.0%, was equal to the premiums paid under the same 3.0% rate, so no additional reserve was needed.

This change has significant implications for the company's operations. Although the higher interest rate assumption leads to lower premium income in the short term, the need to establish and maintain a larger reserve reduces the company's immediate cash flow and liquidity. More funds must be set aside to cover the anticipated future liabilities, which can place pressure on the company's capital reserves. This increased reserve requirement may limit the company's ability to invest in other opportunities, reduce its available working capital, and potentially affect its ability to distribute dividends. The company will need to carefully manage these reserves and balance long-term obligations with operational flexibility to ensure its financial stability and solvency in the future.

A Appendix

This contains excel preview of the tables.

Duration 0 l[x]	Duration 1 l[x-1]+1	Duration 2+ lx	Age x	Duration 0 e[x]	Duration 1 e[x−1]+1	Duration 2+ ex
9997.8091		10000.0000	17	61.353		61.339
9991.8904	9993.5400	9994.0000	18	60.389	60.379	60.376
9986.0351	9987.6338	9988.0636	19	59.424	59.414	59.412
9980.2432	9981.7911	9982.2006	20	58.458	58.449	58.447
9974.5046	9976.0016	9976.3909	21	57.492	57.483	57.481
9968.8391	9970.2654	9970.6346	22	56.524	56.516	56.514
9963.1967	9964.5824	9964.9313	23	55.556	55.548	55.546
9957.5775	9958.9225	9959.2613	24	54.587	54.580	54.578
9951.9913	9953.2858	9953.6144	25	53.618	53.611	53.609
9946.3982	9947.6622	9947.9807	26	52.648	52.641	52.639
9940.7984	9942.0218	9942.3402	27	51.677	51.671	51.669
9935.1818	9936.3549	9936.6730	28	50.706	50.700	50.699
9929.5088	9930.6613	9930.9694	29	49.735	49.729	49.728
9923.7497	9924.8916	9925.2094	30	48.764	48.758	48.757
9917.9145	9919.0260	9919.3535	31	47.792	47.787	47.785
9911.9538	9913.0547	9913.3821	32	46.821	46.816	46.814
9905.8282	9906.9285	9907.2655	33	45.850	45.845	45.843
9899.4984	9900.6078	9900.9645	34	44.879	44.874	44.872
9892.9151	9894.0536	9894.4299	35	43.909	43.904	43.902
9886.0395	9887.2069	9887.6126	36	42.939	42.934	42.932
9878.8128	9880.0288	9880.4540	37	41.970	41.965	41.963
9871.1665	9872.4508	9872.8954	38	41.003	40.997	40.995
9863.0227	9864.4047	9864.8688	39	40.036	40.031	40.029
9854.3036	9855.7931	9856.2863	40	39.071	39.066	39.064
9844.9025	9846.5384	9847.0510	41	38.108	38.102	38.100
9834.7030	9836.5245	9837.0661	42	37.148	37.141	37.139
9823.5994	9825.6354	9826.2060	43	36.189	36.182	36.180
9811.4473	9813.7463	9814.3359	44	35.234	35.226	35.224
9798.0837	9800.6939	9801.3123	45	34.282	34.273	34.271
9783.3371	9786.3162	9786.9534	46	33.333	33.323	33.321
9766.9983	9770.4231	9771.0789	47	32.388	32.377	32.375
9748.8603	9752.7874	9753.4714	48	31.448	31.436	31.433
9728.6499	9733.1938	9733.8865	49	30.513	30.499	30.497
9706.0977	9711.3524	9712.0728	50	29.583	29.567	29.565
9680.8990	9686.9669	9687.7149	51	28.660	28.642	28.639
9652.6965	9659.7075	9660.5021	52	27.742	27.722	27.720
9621.1006	9629.2115	9630.0522	53	26.833	26.810	26.808
9585.6916	9595.0563	9595.9715	54	25.931	25.905	25.903
9545.9929	9556.8003	9557.8179	55	25.037	25.009	25.006
9501.4839	9513.9375	9515.1040	56	24.153	24.122	24.119
9451.5938	9465.9293	9467.2906	57	23.279	23.244	23.240
9395.6971	9412.1712	9413.8004	58	22.415	22.376	22.373
9333.1284	9352.0165	9354.0040	59	21.563	21.520	21.516
9263.1422	9284.7641	9287.2164	60	20.724	20.676	20.670

Figure 4: Life table

9184.9687	9209.6568	9212.7143	61	19.897	19.844	19.837
9097.7405	9125.8818	9129.7170	62	19.084	19.026	19.018
9000.5884	9032.5642	9037.3973	63	18.286	18.222	18.212
8892.5741	8928.8177	8934.8771	64	17.503	17.433	17.421
8772.7359	8813.6881	8821.2612	65	16.736	16.660	16.645
8640.0481	8686.2016	8695.6199	66	15.987	15.903	15.886
8493.5187	8545.3532	8557.0118	67	15.255	15.164	15.143
8332.1396	8390.1611	8404.4916	68	14.541	14.443	14.418
8154.9318	8219.6390	8237.1329	69	13.847	13.740	13.711
7960.9776	8032.8606	8054.0544	70	13.172	13.057	13.023
7749.4659	7828.9686	7854.4508	71	12.517	12.394	12.354
7519.7027	7607.2400	7637.6208	72	11.883	11.751	11.704
7271.1461	7367.0828	7403.0084	73	11.270	11.129	11.075
7003.5216	7108.1052	7150.2401	74	10.679	10.529	10.467
6716.8231	6830.1844	6879.1673	75	10.110	9.950	9.879
6411.3459	6533.5008	6589.9258	76	9.562	9.393	9.313
6087.8084	6218.5759	6282.9803	77	9.037	8.859	8.768
5747.3624	5886.3628	5959.1680	78	8.534	8.346	8.244
5391.6400	5538.2791	5619.7577	79	8.053	7.856	7.742
5022.7931	5176.2224	5266.4604	80	7.594	7.388	7.261
4643.5129	4802.6290	4901.4789	81	7.157	6.942	6.802
4257.0056	4420.4525	4527.4960	82	6.741	6.518	6.364
3866.9884	4033.1467	4147.6708	83	6.347	6.116	5.947
3477.5929	3644.6327	3765.5998	84	5.974	5.734	5.550
3093.2863	3259.1862	3385.2479	85	5.620	5.374	5.174
2718.7128	2881.3467	3010.8395	86	5.287	5.034	4.817
2358.5299	2515.7310	2646.7416	87	4.972	4.713	4.480
2017.2298	2166.8805	2297.2976	88	4.676	4.412	4.161
1698.9089	1839.0458	1966.6499	89	4.397	4.129	3.861
1407.0550	1535.9801	1658.5545	90	4.136	3.864	3.578
	1260.7354	1376.1906	91		3.616	3.312
		1121.9889	92			3.063
		897.5025	93			2.829
		703.3242	94			2.610
		539.0643	95			2.405
		403.4023	96			2.214
		294.2061	97			2.035
		208.7060	98			1.869
		143.7120	99			1.715
		95.8476	100			1.571
		61.7733	101			1.437
		38.3796	102			1.314
		22.9284	103			1.199
		13.1359	104			1.093
		7.1968	105			0.994
		3.7596	106			0.904
		1.8669	107			0.820
		0.8784	108			0.743
		0.3903	109			0.672
		0.1632	110			0.606
		0.0640	111			0.546
		0.0234	112			0.491
		0.0080	113			0.440
		0.0025	114			0.394
		0.0007	115			0.352
		0.0002	116			0.313
		0.0000	117			0.277
		0.0000	118			0.240
		0.0000	119			0.183
		0.0000	120			0.000

Age x	Duration 0 $q[x]$	Duration 1 $q[x-1]+1$	Duration 2+ qx	Duration 0 $l[x]$	Duration 1 $l[x-1]+1$	Duration 2+ lx	Duration 0 $e[x]$	Duration 1 $e[x-1]+1$	Duration 2+ ex
17	0.0003843	0.000497	0.00054	9998.0283		10000	62.41078443		62.39852
18	0.0003834	0.000493	0.0005346	9992.7013	9994.186	9994.6	61.44386732	61.4347781	61.432233
19	0.0003825	0.00049	0.0005283	9987.4312	9988.8701	9989.2569	60.47610892	60.4674339	60.465093
20	0.0003825	0.000487	0.0005238	9982.2177	9983.611	9983.9796	59.50751925	59.4992499	59.497053
21	0.0003825	0.000484	0.0005193	9977.0519	9978.3995	9978.75	58.53816196	58.5302896	58.528234
22	0.0003843	0.000482	0.0005148	9971.9517	9973.2357	9973.568	57.56794172	57.5605614	57.558644
23	0.0003861	0.000481	0.0005121	9966.8719	9968.1194	9968.4336	56.59712602	56.5900736	56.58829
24	0.0003879	0.00048	0.0005103	9961.8128	9963.0237	9963.3288	55.62571805	55.6189866	55.617284
25	0.0003915	0.00048	0.0005094	9956.7830	9957.9486	9958.2445	54.65367183	54.6473036	54.64568
26	0.000396	0.000482	0.0005103	9951.7469	9952.885	9953.1717	53.68118652	53.6750771	53.673531
27	0.0004023	0.000484	0.000513	9946.7043	9947.806	9948.0926	52.70826105	52.7024527	52.700934
28	0.0004095	0.000488	0.0005166	9941.6464	9942.7028	9942.9893	51.73494272	51.7294741	51.727984
29	0.0004185	0.000494	0.000522	9936.5374	9937.5753	9937.8527	50.76140978	50.7561369	50.75472
30	0.0004284	0.000502	0.000531	9931.3505	9932.379	9932.6652	49.78778789	49.7826623	49.781228
31	0.000441	0.000512	0.0005418	9926.0948	9927.0959	9927.3909	48.81401906	48.8091261	48.807676
32	0.0004563	0.000526	0.0005553	9920.7258	9921.7174	9922.0123	47.84030639	47.8355555	47.834134
33	0.0004743	0.000542	0.0005724	9915.2078	9916.199	9916.5026	46.86679773	46.8621459	46.860711
34	0.000495	0.000562	0.000594	9909.5056	9910.5051	9910.8264	45.89363141	45.8890372	45.88755
35	0.0005193	0.000586	0.0006201	9903.5747	9904.6004	9904.9393	44.92097489	44.91636	44.914823
36	0.0005472	0.000615	0.0006516	9897.3800	9898.4318	9898.7973	43.94894577	43.9443145	43.942692
37	0.0005796	0.00065	0.0006885	9890.8684	9891.9641	9892.3472	42.97772776	42.9730078	42.971344
38	0.0006165	0.000691	0.0007317	9883.9784	9885.1357	9885.5363	42.0075278	42.0026521	42.00095
39	0.0006597	0.000741	0.000783	9876.6395	9877.8849	9878.3031	41.03857079	41.0334414	41.031704
40	0.0007092	0.000798	0.0008433	9868.7815	9870.1238	9870.5684	40.07106465	40.0656618	40.063857
41	0.0007659	0.000866	0.0009126	9860.3081	9861.7826	9862.2445	39.10530068	39.0995032	39.097672
42	0.0008298	0.000944	0.0009936	9851.1142	9852.756	9853.2442	38.14157821	38.1352744	38.133385
43	0.0009027	0.001035	0.0010872	9841.1042	9842.9397	9843.4541	37.1801335	37.1732544	37.171312
44	0.0009864	0.00114	0.0011943	9830.1479	9832.2207	9832.7523	36.22130561	36.2137263	36.211769
45	0.0010809	0.001262	0.0013185	9818.0978	9820.4514	9821.009	35.26546305	35.2570696	35.255068
46	0.001188	0.001401	0.0014598	9804.7988	9807.4854	9808.06	34.31296195	34.3036227	34.301613
47	0.0013095	0.001562	0.0016218	9790.0617	9793.1507	9793.7422	33.36423512	33.3537742	33.35176
48	0.0014463	0.001744	0.0018072	9773.6990	9777.2416	9777.8587	32.41966582	32.4079829	32.405938
49	0.0016002	0.001953	0.0020169	9755.4634	9759.5633	9760.1882	31.47977996	31.4666223	31.464608
50	0.0017739	0.002191	0.0022572	9735.1105	9739.8527	9740.5028	30.54503716	30.5302346	30.528197
51	0.0019701	0.002459	0.0025281	9712.3641	9717.8414	9718.5166	29.61593599	29.5993173	29.597261
52	0.0021897	0.002763	0.0028368	9686.8996	9693.2298	9693.9472	28.69305739	28.6743975	28.672275
53	0.0024363	0.003107	0.0031851	9658.3627	9665.6882	9666.4474	27.77699054	27.7560245	27.753844
54	0.0027126	0.003493	0.0035784	9626.3715	9634.8321	9635.6588	26.86832717	26.8448289	26.842526
55	0.0030222	0.003927	0.0040221	9590.4914	9600.259	9601.1786	25.96771887	25.9414084	25.938924
56	0.0033678	0.004413	0.0045225	9550.2471	9561.507	9562.5617	25.07583786	25.0464364	25.043674
57	0.0037539	0.004956	0.005085	9505.1162	9518.0837	9519.315	24.19338002	24.1605736	24.157449
58	0.0041841	0.005562	0.0057168	9454.5252	9469.435	9470.9093	23.32107115	23.2845418	23.280917
59	0.0046638	0.006236	0.006426	9397.8618	9414.9665	9416.766	22.45962639	22.4190588	22.414775
60	0.0051966	0.006984	0.0072198	9334.4388	9354.0321	9356.2538	21.60983271	21.5648644	21.559744

Figure 5: AM92 tables after reduction of 10%

61	0.0057897	0.007812	0.0081081	9263.5431	9285.9315	9288.7035	20.77242464	20.722717	20.716533
62	0.0064476	0.008726	0.0091008	9184.3684	9209.9099	9213.3898	19.94823627	19.8933911	19.885877
63	0.0071766	0.009734	0.0102096	9096.1021	9125.1513	9129.5406	19.13800986	19.0776892	19.068517
64	0.0079839	0.010841	0.0114444	8997.8616	9030.823	9036.3316	18.34253326	18.2763485	18.265207
65	0.0088776	0.012056	0.0128187	8888.7353	8926.0236	8932.9162	17.56256248	17.4901568	17.476661
66	0.009864	0.013386	0.014346	8767.7437	8809.8247	8818.4079	16.79890616	16.7198724	16.703598
67	0.0109521	0.014836	0.0160416	8633.9265	8681.2587	8691.899	16.05227307	15.9662614	15.946716
68	0.0121518	0.016415	0.0179217	8486.2947	8539.3668	8552.467	15.32338453	15.2300259	15.206697
69	0.0134721	0.018131	0.0200034	8323.8702	8383.1709	8399.1923	14.61293053	14.5118818	14.484201
70	0.0149238	0.019989	0.0223047	8145.7117	8211.7301	8231.1799	13.9215542	13.8124858	13.779848
71	0.0165177	0.021997	0.0248454	7950.9545	8024.1467	8047.5859	13.24983094	13.1324643	13.094215
72	0.0182664	0.024162	0.0276462	7738.8166	7819.623	7847.6404	12.59830259	12.4723634	12.427835
73	0.0201807	0.026491	0.0307296	7508.6288	7597.4563	7630.6829	11.96748488	11.83271	11.781186
74	0.022275	0.028987	0.0341199	7259.9384	7357.0994	7396.1951	11.35776686	11.2139714	11.154695
75	0.0245637	0.031658	0.0378414	6992.5093	7098.2232	7143.8377	10.7694825	10.616525	10.548737
76	0.0270603	0.03451	0.0419202	6706.3518	6820.7474	6873.5049	10.20293466	10.0406825	9.9636155
77	0.029781	0.037544	0.0463842	6401.8369	6524.8759	6585.3662	9.658300999	9.48670813	9.3995674
78	0.0327411	0.040763	0.0512604	6079.7098	6211.1838	6279.9092	9.135689117	8.95476382	8.8567656
79	0.0359586	0.044172	0.0565803	5741.1398	5880.6534	5957.9986	8.635129978	8.4449264	8.3352967
80	0.0394497	0.04777	0.0623727	5387.7673	5534.6965	5620.8932	8.156571781	7.95721903	7.8351946
81	0.0432333	0.051559	0.06867	5021.7357	5175.2215	5270.3029	7.699873048	7.49156133	7.3564062
82	0.0473274	0.055538	0.0755037	4645.6870	4804.6295	4908.3912	7.264830218	7.04780627	6.8988181
83	0.0517509	0.059703	0.0829053	4262.7814	4425.8187	4537.7895	6.851138127	6.6257365	6.4622452
84	0.0565236	0.064052	0.0909063	3876.6321	4042.1787	4161.5827	6.45843287	6.22504047	6.0464317
85	0.0616644	0.068579	0.09954	3491.2614	3657.5109	3783.2686	6.08626725	5.8453571	5.6510545
86	0.0671949	0.07328	0.1088361	3110.9813	3275.9749	3406.6821	5.734139958	5.48623717	5.2757418
87	0.0731322	0.078144	0.1188252	2740.2665	2901.9392	3035.9121	5.401503238	5.14720048	4.9200578
88	0.0794979	0.083165	0.1295361	2383.6121	2539.8648	2675.1692	5.087730278	4.82769543	4.5835208
89	0.0863118	0.08833	0.140994	2045.3286	2194.12	2328.6382	4.792156128	4.52712512	4.265607
90	0.093591	0.093628	0.1532223	1729.3420	1868.7926	2000.3142	4.514093193	4.24484844	3.9657476
91		0.099047	0.1662426		1567.4912	1693.8215		3.98019458	3.6833397
92			0.1800711			1412.2362			3.4177595
93			0.1947186			1157.9333			3.1683609
94			0.2101932			932.46212			2.9344766
95			0.2264958			736.46492			2.715436
96			0.2436192			569.65871			2.510564
97			0.2615517			430.87891			2.31918
98			0.2802726			318.1818			2.1406126
99			0.2997522			229.00416			1.9741991
100			0.3199545			160.35966			1.8192864
101			0.3408318			109.05186			1.6752421
102			0.3623292			71.883521			1.5414486
103			0.384381			45.838022			1.417311
104			0.4069143			28.218757			1.3022534
105			0.4298472			16.736142			1.1957256
106			0.4530888			9.542158			1.097202
107			0.4765437			5.2187131			1.0061794
108			0.5001066			2.7317682			0.9221841
109			0.5236713			1.3655929			0.8447615
110			0.5471262			0.6504711			0.7734845
111			0.5703579			0.2945813			0.707947
112			0.5932539			0.1265645			0.6477599
113			0.6157026			0.0514796			0.5925411
114			0.6375978			0.0197835			0.5418816
115			0.6588378			0.0071696			0.4952492
116			0.6793281			0.002446			0.4516532
117			0.6989832			0.0007844			0.4084589
118			0.7177293			0.0002361			0.3569306
119			0.7355025			6.665E-05			0.2644975
120			0.9			1.763E-05			0

Age x	Duration 0 $D[x]$	Duration 1 $D[x-1]+1$	Duration 2+ Dx	Duration 0 $d[x]$	Duration 1 $d[x-1]+1$	Duration 2+ dx	Duration 0 $c[x]$	Duration 1 $c[x-1]+1$	Duration 2+ cx
17	5132.61		5133.73	4.2691		6.0000	2.11		2.96
18	4932.28	4933.09	4933.32	4.2565	5.4765	5.9364	2.02	2.60	2.82
19	4739.80	4740.55	4740.76	4.2441	5.4333	5.8630	1.94	2.48	2.68
20	4554.85	4555.56	4555.75	4.2416	5.4001	5.8096	1.86	2.37	2.55
21	4377.15	4377.80	4377.98	4.2392	5.3671	5.7564	1.79	2.26	2.43
22	4206.41	4207.01	4207.16	4.2567	5.3341	5.7032	1.73	2.16	2.31
23	4042.33	4042.89	4043.04	4.2742	5.3211	5.6700	1.67	2.08	2.21
24	3884.66	3885.19	3885.32	4.2917	5.3081	5.6469	1.61	1.99	2.12
25	3733.16	3733.64	3733.77	4.3291	5.3051	5.6337	1.56	1.91	2.03
26	3587.56	3588.01	3588.13	4.3764	5.3220	5.6405	1.52	1.85	1.96
27	3447.63	3448.06	3448.17	4.4435	5.3488	5.6671	1.48	1.78	1.89
28	3313.16	3313.55	3313.66	4.5205	5.3855	5.7037	1.45	1.73	1.83
29	3183.91	3184.28	3184.38	4.6172	5.4519	5.7600	1.42	1.68	1.78
30	3059.68	3060.03	3060.13	4.7237	5.5381	5.8559	1.40	1.64	1.74
31	2940.27	2940.60	2940.69	4.8598	5.6439	5.9715	1.39	1.61	1.70
32	2825.48	2825.79	2825.89	5.0254	5.7892	6.1166	1.38	1.59	1.68
33	2715.13	2715.43	2715.52	5.2204	5.9640	6.3010	1.38	1.57	1.66
34	2609.03	2609.33	2609.42	5.4447	6.1780	6.5346	1.38	1.57	1.66
35	2507.02	2507.31	2507.40	5.7082	6.4410	6.8173	1.39	1.57	1.66
36	2408.92	2409.20	2409.30	6.0107	6.7530	7.1586	1.41	1.58	1.68
37	2314.57	2314.86	2314.96	6.3620	7.1334	7.5585	1.43	1.61	1.70
38	2223.83	2224.12	2224.22	6.7617	7.5820	8.0267	1.46	1.64	1.74
39	2136.53	2136.83	2136.93	7.2296	8.1184	8.5824	1.51	1.69	1.79
40	2052.54	2052.85	2052.96	7.7652	8.7421	9.2353	1.56	1.75	1.85
41	1971.72	1972.04	1972.15	8.3780	9.4724	9.9849	1.61	1.82	1.92
42	1893.92	1894.27	1894.37	9.0676	10.3185	10.8601	1.68	1.91	2.01
43	1819.02	1819.40	1819.50	9.8531	11.2995	11.8701	1.75	2.01	2.11
44	1746.89	1747.30	1747.41	10.7533	12.4340	13.0236	1.84	2.13	2.23
45	1677.42	1677.86	1677.97	11.7675	13.7406	14.3589	1.94	2.26	2.36
46	1610.47	1610.96	1611.07	12.9140	15.2373	15.8744	2.04	2.41	2.51
47	1545.95	1546.49	1546.59	14.2110	16.9517	17.6075	2.16	2.58	2.68
48	1483.73	1484.32	1484.43	15.6664	18.9009	19.5850	2.29	2.77	2.87
49	1423.70	1424.37	1424.47	17.2975	21.1210	21.8136	2.43	2.97	3.07
50	1365.77	1366.51	1366.61	19.1307	23.6374	24.3579	2.59	3.20	3.30
51	1309.83	1310.65	1310.75	21.1915	26.4648	27.2128	2.76	3.44	3.54
52	1255.78	1256.70	1256.80	23.4850	29.6553	30.4499	2.94	3.71	3.81
53	1203.53	1204.55	1204.65	26.0443	33.2400	34.0808	3.13	4.00	4.10
54	1152.98	1154.11	1154.22	28.8913	37.2384	38.1536	3.34	4.31	4.41
55	1104.05	1105.30	1105.41	32.0554	41.6963	42.7139	3.56	4.64	4.75
56	1056.63	1058.02	1058.15	35.5546	46.6468	47.8134	3.80	4.99	5.11
57	1010.66	1012.19	1012.34	39.4226	52.1289	53.4902	4.05	5.36	5.50
58	966.04	967.73	967.90	43.6806	58.1672	59.7965	4.32	5.75	5.91
59	922.70	924.57	924.76	48.3643	64.8001	66.7876	4.60	6.16	6.35
60	880.56	882.61	882.85	53.4854	72.0498	74.5020	4.89	6.59	6.81

Figure 6: Computational fuctions at 4%

61	839.55	841.80	842.08	59.0869	79.9398	82.9973	5.19	7.03	7.29
62	799.59	802.06	802.40	65.1762	88.4846	92.3197	5.51	7.48	7.80
63	760.62	763.33	763.74	71.7707	97.6872	102.5202	5.83	7.94	8.33
64	722.59	725.54	726.03	78.8860	107.5565	113.6159	6.16	8.40	8.88
65	685.44	688.64	689.23	86.5343	118.0682	125.6412	6.50	8.87	9.44
66	649.11	652.57	653.28	94.6949	129.1899	138.6082	6.84	9.33	10.01
67	613.56	617.30	618.14	103.3576	140.8616	152.5202	7.18	9.78	10.59
68	578.75	582.78	583.77	112.5005	153.0281	167.3586	7.51	10.22	11.18
69	544.65	548.97	550.14	122.0712	165.5846	183.0785	7.84	10.63	11.76
70	511.25	515.87	517.23	132.0089	178.4098	199.6036	8.15	11.02	12.33
71	478.53	483.43	485.01	142.2259	191.3478	216.8300	8.44	11.36	12.87
72	446.48	451.68	453.48	152.6199	204.2316	234.6124	8.71	11.66	13.39
73	415.12	420.59	422.64	163.0409	216.8427	252.7683	8.95	11.90	13.88
74	384.46	390.20	392.51	173.3372	228.9379	271.0728	9.15	12.08	14.31
75	354.54	360.52	363.11	183.3223	240.2586	289.2415	9.30	12.19	14.68
76	325.40	331.60	334.46	192.7699	250.5206	306.9456	9.41	12.23	14.98
77	297.09	303.48	306.62	201.4456	259.4079	323.8122	9.45	12.17	15.19
78	269.69	276.21	279.63	209.0833	266.6051	339.4104	9.43	12.03	15.31
79	243.27	249.89	253.56	215.4176	271.8187	353.2973	9.35	11.79	15.33
80	217.91	224.57	228.48	220.1641	274.7435	364.9815	9.18	11.46	15.23
81	193.71	200.35	204.47	223.0604	275.1330	373.9828	8.95	11.04	15.00
82	170.75	177.31	181.60	223.8589	272.7817	379.8252	8.63	10.52	14.65
83	149.14	155.55	159.97	222.3557	267.5468	382.0710	8.25	9.92	14.17
84	128.97	135.16	139.65	218.4067	259.3849	380.3519	7.79	9.25	13.56
85	110.30	116.22	120.71	211.9396	248.3467	374.4084	7.27	8.52	12.84
86	93.22	98.79	103.23	202.9818	234.6050	364.0978	6.69	7.73	12.00
87	77.76	82.94	87.26	191.6494	218.4334	349.4440	6.08	6.92	11.08
88	63.95	68.69	72.83	178.1839	200.2306	330.6478	5.43	6.10	10.08
89	51.78	56.06	59.95	162.9288	180.4913	308.0954	4.78	5.29	9.03
90	41.24	45.02	48.61	146.3197	159.7895	282.3639	4.12	4.50	7.96
91		35.53	38.78		138.7464	254.2017		3.76	6.89
92			30.40			224.4864			5.85
93			23.38			194.1783			4.86
94			17.62			164.2600			3.96
95			12.99			135.6620			3.14
96			9.34			109.1962			2.43
97			6.55			85.5001			1.83
98			4.47			64.9940			1.34
99			2.96			47.8644			0.95
100			1.90			34.0743			0.65
101			1.18			23.3937			0.43
102			0.70			15.4512			0.27
103			0.40			9.7925			0.17
104			0.22			5.9391			0.10
105			0.12			3.4373			0.05
106			0.06			1.8927			0.03
107			0.03			0.9885			0.01
108			0.01			0.4881			0.01
109			0.01			0.2271			0.00
110			0.00			0.0992			0.00
111			0.00			0.0405			
112			0.00			0.0154			
113			0.00			0.0055			
114			0.00			0.0018			
115			0.00			0.0005			
116			0.00			0.0001			
117			0.00			0.0000			
118			0.00			0.0000			
119			0.00			0.0000			
120			0.00			0.0000			

Age x	Duration 0 R[x]	Duration 1 R[x−1]+1	Duration 2+ Rx
17	27724.52		27725.81
18	27204.73	27205.71	27205.92
19	26687.90	26688.80	26689.00
20	26173.87	26174.71	26174.89
21	25662.51	25663.29	25663.45
22	25153.71	25154.42	25154.57
23	24647.31	24647.98	24648.12
24	24143.23	24143.85	24143.97
25	23641.35	23641.93	23642.04
26	23141.58	23142.12	23142.23
27	22643.84	22644.35	22644.45
28	22148.06	22148.53	22148.63
29	21654.16	21654.60	21654.70
30	21162.07	21162.50	21162.60
31	20671.77	20672.17	20672.27
32	20183.20	20183.58	20183.68
33	19696.32	19696.70	19696.79
34	19211.11	19211.48	19211.57
35	18727.56	18727.93	18728.02
36	18245.66	18246.03	18246.12
37	17765.42	17765.79	17765.89
38	17286.86	17287.23	17287.33
39	16809.99	16810.38	16810.47
40	16334.86	16335.26	16335.36
41	15861.52	15861.93	15862.03
42	15390.01	15390.45	15390.55
43	14920.43	14920.89	14920.99
44	14452.85	14453.34	14453.45
45	13987.38	13987.91	13988.01
46	13524.14	13524.71	13524.81
47	13063.25	13063.87	13063.97
48	12604.88	12605.55	12605.65
49	12149.17	12149.90	12150.00
50	11696.32	11697.12	11697.22

Figure 7: Computational fuctions Rx at 4%

51	11246.52	11247.41	11247.51
52	10800.02	10800.99	10801.09
53	10357.04	10358.12	10358.22
54	9917.85	9919.05	9919.15
55	9482.75	9484.07	9484.19
56	9052.04	9053.51	9053.63
57	8626.06	8627.69	8627.83
58	8205.16	8206.97	8207.14
59	7789.74	7791.76	7791.95
60	7380.21	7382.44	7382.67
61	6976.98	6979.46	6979.73
62	6580.53	6583.29	6583.61
63	6191.34	6194.39	6194.79
64	5809.91	5813.29	5813.76
65	5436.77	5440.50	5441.07
66	5072.46	5076.57	5077.25
67	4717.54	4722.06	4722.87
68	4372.59	4377.55	4378.51
69	4038.20	4043.61	4044.74
70	3714.94	3720.83	3722.14
71	3403.41	3409.79	3411.31
72	3104.17	3111.06	3112.79
73	2817.78	2825.18	2827.16
74	2544.78	2552.69	2554.91
75	2285.66	2294.06	2296.55
76	2040.87	2049.73	2052.49
77	1810.80	1820.09	1823.11
78	1595.76	1605.43	1608.71
79	1395.99	1405.97	1409.51
80	1211.64	1221.85	1225.62
81	1042.74	1053.09	1057.05
82	889.20	899.59	903.72
83	750.83	761.13	765.38
84	627.27	637.38	641.69
85	518.06	527.85	532.17
86	422.60	431.95	436.22
87	340.15	348.95	353.10
88	269.86	278.01	281.99
89	210.78	218.21	221.95
90	161.90	168.54	171.99
91		127.93	131.06
92			98.09
93			72.01
94			51.78
95			36.41
96			25.00
97			16.73
98			10.89
99			6.89
100			4.22
101			2.50
102			1.43
103			0.79
104			0.41
105			0.21
106			0.10
107			0.05
108			0.02
109			0.01
110			0.00

Duration 0 N[x]	Duration 1 N[x−1]+1	Duration 2+ Nx	Age x	Duration 0 M[x]	Duration 1 M[x−1]+1	Duration 2+ Mx	Duration 0 S[x]	Duration 1 S[x−1]+1	Duration 2+ Sx
119958.58		119959.94	17	518.82		519.89	2398085.62		2398087.19
114824.96	114825.98	114826.20	18	515.93	516.71	516.93	2278125.81	2278127.03	2278127.26
109891.73	109892.68	109892.88	19	513.19	513.91	514.11	2163299.72	2163300.85	2163301.06
105151.06	105151.94	105152.13	20	510.58	511.25	511.43	2053406.94	2053407.99	2053408.17
100595.40	100596.21	100596.38	21	508.09	508.72	508.88	1948254.91	1948255.88	1948256.05
96217.50	96218.25	96218.40	22	505.73	506.30	506.45	1847658.63	1847659.51	1847659.67
92010.40	92011.10	92011.24	23	503.47	504.00	504.14	1751440.30	1751441.12	1751441.26
87967.43	87968.07	87968.21	24	501.30	501.80	501.93	1659429.12	1659429.89	1659430.02
84082.16	84082.76	84082.88	25	499.23	499.69	499.81	1571460.98	1571461.69	1571461.82
80348.43	80349.00	80349.12	26	497.23	497.67	497.78	1487378.14	1487378.82	1487378.93
76760.35	76760.88	76760.99	27	495.31	495.72	495.82	1407029.07	1407029.71	1407029.82
73312.22	73312.71	73312.82	28	493.46	493.83	493.93	1330268.14	1330268.73	1330268.83
69998.60	69999.06	69999.16	29	491.66	492.01	492.10	1256955.35	1256955.92	1256956.01
66814.23	66814.68	66814.78	30	489.90	490.23	490.33	1186956.21	1186956.75	1186956.85
63754.13	63754.56	63754.65	31	488.18	488.50	488.59	1120141.46	1120141.97	1120142.07
60813.46	60813.87	60813.96	32	486.50	486.80	486.89	1056386.82	1056387.32	1056387.42
57987.58	57987.98	57988.07	33	484.84	485.12	485.21	995572.87	995573.36	995573.46
55272.07	55272.45	55272.55	34	483.18	483.46	483.55	937584.81	937585.29	937585.38
52662.65	52663.03	52663.13	35	481.53	481.80	481.90	882312.25	882312.74	882312.83
50155.24	50155.63	50155.73	36	479.87	480.14	480.23	829649.12	829649.61	829649.71
47745.94	47746.33	47746.43	37	478.19	478.46	478.56	779493.39	779493.88	779493.98
45430.98	45431.37	45431.47	38	476.48	476.76	476.85	731746.96	731747.45	731747.55
43206.74	43207.15	43207.25	39	474.74	475.02	475.12	686315.48	686315.99	686316.09
41069.80	41070.21	41070.31	40	472.94	473.23	473.33	643108.22	643108.74	643108.84
39016.82	39017.25	39017.36	41	471.07	471.38	471.48	602037.89	602038.43	602038.53
37044.65	37045.10	37045.21	42	469.12	469.46	469.56	563020.51	563021.07	563021.17
35150.25	35150.73	35150.84	43	467.08	467.44	467.54	525975.27	525975.86	525975.96
33330.72	33331.23	33331.34	44	464.94	465.33	465.43	490824.40	490825.02	490825.13
31583.27	31583.82	31583.93	45	462.67	463.10	463.20	457493.03	457493.68	457493.79
29905.26	29905.86	29905.96	46	460.27	460.74	460.84	425909.06	425909.76	425909.86
28294.14	28294.79	28294.89	47	457.71	458.23	458.33	396003.05	396003.80	396003.90
26747.50	26748.20	26748.30	48	454.97	455.55	455.65	367708.10	367708.90	367709.01
25263.01	25263.77	25263.87	49	452.05	452.68	452.78	340959.74	340960.61	340960.71
23838.46	23839.30	23839.41	50	448.91	449.61	449.71	315695.79	315696.73	315696.83
22471.77	22472.69	22472.79	51	445.53	446.32	446.41	291856.30	291857.33	291857.43
21160.92	21161.94	21162.04	52	441.90	442.77	442.87	269383.41	269384.53	269384.63
19904.01	19905.14	19905.24	53	437.99	438.96	439.07	248221.25	248222.49	248222.59
18699.23	18700.48	18700.59	54	433.78	434.86	434.97	228315.88	228317.24	228317.35
17544.87	17546.25	17546.37	55	429.24	430.44	430.55	209615.14	209616.65	209616.76
16439.29	16440.82	16440.95	56	424.35	425.68	425.80	192068.59	192070.26	192070.39
15380.96	15382.66	15382.81	57	419.08	420.55	420.69	175627.43	175629.29	175629.44
14368.41	14370.30	14370.47	58	413.41	415.03	415.19	160244.38	160246.47	160246.63
13400.27	13402.37	13402.57	59	407.30	409.09	409.28	145873.63	145875.97	145876.16
12475.24	12477.57	12477.80	60	400.74	402.71	402.93	132470.75	132473.36	132473.60

Figure 8: Computational fuctions at 4%

11592.08	11594.68	11594.96	61	393.69	395.85	396.12	119992.58	119995.51	119995.79
10749.66	10752.54	10752.88	62	386.14	388.50	388.83	108397.21	108400.50	108400.84
9946.87	9950.07	9950.48	63	378.05	380.63	381.02	97643.87	97647.55	97647.96
9182.71	9186.25	9186.74	64	369.41	372.22	372.69	87692.86	87696.99	87697.48
8456.21	8460.12	8460.71	65	360.20	363.25	363.82	78505.54	78510.15	78510.74
7766.46	7770.77	7771.48	66	350.40	353.70	354.38	70044.17	70049.32	70050.03
7112.62	7117.36	7118.20	67	339.99	343.55	344.36	62271.97	62277.71	62278.55
6493.86	6499.06	6500.06	68	328.98	332.81	333.77	55152.99	55159.35	55160.35
5909.43	5915.12	5916.29	69	317.37	321.47	322.59	48652.08	48659.12	48660.29
5358.59	5364.78	5366.14	70	305.15	309.53	310.84	42734.88	42742.64	42744.00
4840.63	4847.34	4848.92	71	292.35	297.00	298.51	37367.77	37376.29	37377.86
4354.86	4362.10	4363.91	72	278.98	283.90	285.64	32517.84	32527.14	32528.95
3900.59	3908.38	3910.43	73	265.09	270.27	272.24	28152.88	28162.99	28165.04
3477.14	3485.47	3487.78	74	250.72	256.14	258.37	24241.38	24252.30	24254.61
3083.84	3092.69	3095.27	75	235.93	241.57	244.06	20752.53	20764.24	20766.83
2719.96	2729.30	2732.16	76	220.78	226.62	229.38	17656.20	17668.69	17671.55
2384.76	2394.56	2397.70	77	205.37	211.38	214.40	14923.03	14936.25	14939.39
2077.47	2087.67	2091.08	78	189.79	195.92	199.20	12524.40	12538.27	12541.69
1797.25	1807.78	1811.45	79	174.14	180.35	183.89	10432.48	10446.93	10450.60
1543.20	1553.98	1557.89	80	158.56	164.80	168.56	8620.33	8635.24	8639.15
1314.35	1325.29	1329.41	81	143.15	149.37	153.34	7061.91	7077.14	7081.26
1109.67	1120.65	1124.94	82	128.07	134.21	138.34	5732.17	5747.56	5751.85
928.03	938.92	943.34	83	113.45	119.44	123.69	4607.11	4622.49	4626.91
768.19	778.88	783.37	84	99.42	105.20	109.52	3663.90	3679.09	3683.57
628.87	639.22	643.72	85	86.11	91.63	95.95	2880.91	2895.71	2900.20
508.67	518.57	523.01	86	73.65	78.85	83.12	2237.83	2252.04	2256.48
406.14	415.45	419.77	87	62.14	66.96	71.11	1715.71	1729.16	1733.48
319.75	328.38	332.51	88	51.65	56.06	60.04	1297.05	1309.57	1313.71
247.93	255.80	259.69	89	42.25	46.22	49.96	965.85	977.30	981.19
189.12	196.15	199.74	90	33.96	37.47	40.93	707.63	717.91	721.50
	147.88	151.13	91		29.84	32.97		518.51	521.76
		112.35	92			26.08			370.63
		81.95	93			20.23			258.28
		58.56	94			15.37			176.34
		40.94	95			11.41			117.78
		27.95	96			8.27			76.84
		18.61	97			5.84			48.88
		12.06	98			4.01			30.27
		7.59	99			2.67			18.22
		4.63	100			1.72			10.63
		2.73	101			1.07			6.00
		1.55	102			0.64			3.27
		0.85	103			0.37			1.72
		0.45	104			0.20			0.87
		0.23	105			0.11			0.42
		0.11	106			0.05			0.19
		0.05	107			0.03			0.08
		0.02	108			0.01			0.03
		0.01	109			0.01			0.01
		0.00	110			0.00			0.00

Age x	Duration 0 $D[x]$	Duration 1 $D[x-1]+1$	Duration 2+ Dx	Duration 0 $d[x]$	Duration 1 $d[x-1]+1$	Duration 2+ dx	Duration 0 $c[x]$	Duration 1 $c[x-1]+1$	Duration 2+ cx
17	6048.84		6050.16	4.2691		6.0000	2.51		3.52
18	5869.18	5870.15	5870.42	4.2565	5.4765	5.9364	2.43	3.12	3.39
19	5694.90	5695.81	5696.05	4.2441	5.4333	5.8630	2.35	3.01	3.25
20	5525.82	5526.68	5526.90	4.2416	5.4001	5.8096	2.28	2.90	3.12
21	5361.79	5362.59	5362.80	4.2392	5.3671	5.7564	2.21	2.80	3.00
22	5202.66	5203.41	5203.60	4.2567	5.3341	5.7032	2.16	2.70	2.89
23	5048.27	5048.97	5049.15	4.2742	5.3211	5.6700	2.10	2.62	2.79
24	4898.47	4899.13	4899.30	4.2917	5.3081	5.6469	2.05	2.54	2.70
25	4753.13	4753.74	4753.90	4.3291	5.3051	5.6337	2.01	2.46	2.61
26	4612.09	4612.68	4612.83	4.3764	5.3220	5.6405	1.97	2.40	2.54
27	4475.24	4475.79	4475.93	4.4435	5.3488	5.6671	1.94	2.34	2.48
28	4342.44	4342.95	4343.09	4.5205	5.3855	5.7037	1.92	2.29	2.42
29	4213.55	4214.04	4214.17	4.6172	5.4519	5.7600	1.90	2.25	2.37
30	4088.45	4088.92	4089.05	4.7237	5.5381	5.8559	1.89	2.22	2.34
31	3967.04	3967.48	3967.61	4.8598	5.6439	5.9715	1.89	2.19	2.32
32	3849.18	3849.61	3849.73	5.0254	5.7892	6.1166	1.89	2.18	2.31
33	3734.76	3735.17	3735.30	5.2204	5.9640	6.3010	1.91	2.18	2.31
34	3623.66	3624.07	3624.20	5.4447	6.1780	6.5346	1.93	2.20	2.32
35	3515.78	3516.18	3516.32	5.7082	6.4410	6.8173	1.97	2.22	2.35
36	3411.00	3411.41	3411.55	6.0107	6.7530	7.1586	2.01	2.26	2.40
37	3309.23	3309.64	3309.78	6.3620	7.1334	7.5585	2.07	2.32	2.46
38	3210.36	3210.78	3210.92	6.7617	7.5820	8.0267	2.14	2.39	2.53
39	3114.28	3114.72	3114.87	7.2296	8.1184	8.5824	2.22	2.49	2.63
40	3020.90	3021.36	3021.51	7.7652	8.7421	9.2353	2.31	2.60	2.75
41	2930.12	2930.61	2930.76	8.3780	9.4724	9.9849	2.42	2.74	2.89
42	2841.83	2842.35	2842.51	9.0676	10.3185	10.8601	2.54	2.89	3.05
43	2755.94	2756.51	2756.67	9.8531	11.2995	11.8701	2.68	3.08	3.23
44	2672.36	2672.99	2673.15	10.7533	12.4340	13.0236	2.84	3.29	3.44
45	2590.99	2591.68	2591.85	11.7675	13.7406	14.3589	3.02	3.53	3.69
46	2511.74	2512.50	2512.67	12.9140	15.2373	15.8744	3.22	3.80	3.96
47	2434.51	2435.36	2435.53	14.2110	16.9517	17.6075	3.44	4.10	4.26
48	2359.21	2360.16	2360.33	15.6664	18.9009	19.5850	3.68	4.44	4.60
49	2285.75	2286.82	2286.98	17.2975	21.1210	21.8136	3.95	4.82	4.98
50	2214.03	2215.23	2215.39	19.1307	23.6374	24.3579	4.24	5.23	5.39
51	2143.96	2145.31	2145.47	21.1915	26.4648	27.2128	4.56	5.69	5.85
52	2075.45	2076.96	2077.13	23.4850	29.6553	30.4499	4.90	6.19	6.36
53	2008.41	2010.10	2010.28	26.0443	33.2400	34.0808	5.28	6.74	6.91
54	1942.73	1944.63	1944.82	28.8913	37.2384	38.1536	5.68	7.33	7.51
55	1878.34	1880.46	1880.66	32.0554	41.6963	42.7139	6.12	7.97	8.16
56	1815.13	1817.51	1817.73	35.5546	46.6468	47.8134	6.59	8.65	8.87
57	1753.01	1755.66	1755.92	39.4226	52.1289	53.4902	7.10	9.39	9.63
58	1691.88	1694.85	1695.14	43.6806	58.1672	59.7965	7.64	10.17	10.45
59	1631.66	1634.97	1635.31	48.3643	64.8001	66.7876	8.21	11.00	11.34
60	1572.26	1575.93	1576.35	53.4854	72.0498	74.5020	8.81	11.87	12.28

Figure 9: Computational fuctions at 3%

60	1572.26	1575.93	1576.35	53.4854	72.0498	74.5020	8.81	11.87	12.28
61	1513.59	1517.65	1518.16	59.0869	79.9398	82.9973	9.45	12.79	13.28
62	1455.54	1460.05	1460.66	65.1762	88.4846	92.3197	10.12	13.74	14.34
63	1398.06	1403.03	1403.78	71.7707	97.6872	102.5202	10.82	14.73	15.46
64	1341.05	1346.52	1347.43	78.8860	107.5565	113.6159	11.55	15.75	16.63
65	1284.44	1290.44	1291.55	86.5343	118.0682	125.6412	12.30	16.78	17.86
66	1228.17	1234.73	1236.07	94.6949	129.1899	138.6082	13.07	17.83	19.13
67	1172.18	1179.33	1180.94	103.3576	140.8616	152.5202	13.85	18.87	20.44
68	1116.41	1124.19	1126.11	112.5005	153.0281	167.3586	14.63	19.91	21.77
69	1060.84	1069.26	1071.54	122.0712	165.5846	183.0785	15.42	20.91	23.12
70	1005.45	1014.53	1017.21	132.0089	178.4098	199.6036	16.19	21.88	24.48
71	950.23	959.98	963.10	142.2259	191.3478	216.8300	16.93	22.78	25.81
72	895.20	905.62	909.24	152.6199	204.2316	234.6124	17.64	23.61	27.12
73	840.40	851.49	855.64	163.0409	216.8427	252.7683	18.30	24.33	28.36
74	785.89	797.63	802.35	173.3372	228.9379	271.0728	18.88	24.94	29.53
75	731.77	744.12	749.45	183.3223	240.2586	289.2415	19.39	25.41	30.59
76	678.14	691.06	697.03	192.7699	250.5206	306.9456	19.80	25.73	31.52
77	625.17	638.59	645.21	201.4456	259.4079	323.8122	20.08	25.86	32.28
78	573.01	586.87	594.13	209.0833	266.6051	339.4104	20.24	25.81	32.85
79	521.89	536.09	543.97	215.4176	271.8187	353.2973	20.24	25.54	33.20
80	472.03	486.45	494.93	220.1641	274.7435	364.9815	20.09	25.07	33.30
81	423.67	438.19	447.21	223.0604	275.1330	373.9828	19.76	24.37	33.13
82	377.10	391.57	401.06	223.8589	272.7817	379.8252	19.25	23.46	32.67
83	332.57	346.86	356.71	222.3557	267.5468	382.0710	18.57	22.34	31.90
84	290.37	304.32	314.42	218.4067	259.3849	380.3519	17.71	21.03	30.83
85	250.76	264.21	274.43	211.9396	248.3467	374.4084	16.68	19.55	29.47
86	213.97	226.77	236.97	202.9818	234.6050	364.0978	15.51	17.93	27.82
87	180.22	192.23	202.24	191.6494	218.4334	349.4440	14.22	16.20	25.92
88	149.65	160.75	170.43	178.1839	200.2306	330.6478	12.83	14.42	23.82
89	122.36	132.46	141.65	162.9288	180.4913	308.0954	11.39	12.62	21.54
90	98.39	107.41	115.98	146.3197	159.7895	282.3639	9.93	10.85	19.17
91		85.59	93.43		138.7464	254.2017		9.15	16.76
92			73.95			224.4864			14.37
93			57.43			194.1783			12.06
94			43.70			164.2600			9.91
95			32.52			135.6620			7.94
96			23.62			109.1962			6.21
97			16.73			85.5001			4.72
98			11.52			64.9940			3.48
99			7.70			47.8644			2.49
100			4.99			34.0743			1.72
101			3.12			23.3937			1.15
102			1.88			15.4512			0.74
103			1.09			9.7925			0.45
104			0.61			5.9391			0.27
105			0.32			3.4373			0.15
106			0.16			1.8927			0.08
107			0.08			0.9885			0.04
108			0.04			0.4881			0.02
109			0.02			0.2271			0.01
110			0.01			0.0992			0.00
111			0.00			0.0405			
112			0.00			0.0154			
113			0.00			0.0055			
114			0.00			0.0018			
115			0.00			0.0005			
116			0.00			0.0001			
117			0.00			0.0000			
118			0.00			0.0000			
119			0.00			0.0000			
120			0.00			0.0000			

Duration 0 N[x]	Duration 1 N[x−1]+1	Duration 2+ Nx	Age x	Duration 0 M[x]	Duration 1 M[x−1]+1	Duration 2+ Mx	Duration 0 S[x]	Duration 1 S[x−1]+1	Duration 2+ Sx
171994.71		171996.31	17	1039.28		1040.56	3896874.70		3896876.57
165944.66	165945.87	165946.14	18	1035.84	1036.77	1037.04	3724878.53	3724879.99	3724880.26
160074.34	160075.48	160075.72	19	1032.53	1033.41	1033.65	3558932.51	3558933.87	3558934.12
154378.37	154379.44	154379.67	20	1029.36	1030.18	1030.40	3398856.90	3398858.17	3398858.40
148851.56	148852.56	148852.77	21	1026.30	1027.08	1027.28	3244477.33	3244478.52	3244478.73
143488.85	143489.77	143489.96	22	1023.37	1024.09	1024.28	3095624.68	3095625.77	3095625.97
138285.32	138286.19	138286.36	23	1020.54	1021.22	1021.39	2952134.79	2952135.83	2952136.00
133236.23	133237.05	133237.22	24	1017.80	1018.44	1018.60	2813848.50	2813849.47	2813849.64
128337.00	128337.76	128337.92	25	1015.15	1015.75	1015.90	2680611.35	2680612.27	2680612.42
123583.14	123583.87	123584.02	26	1012.58	1013.15	1013.29	2552273.48	2552274.36	2552274.50
118970.36	118971.05	118971.19	27	1010.08	1010.61	1010.75	2428689.51	2428690.34	2428690.49
114494.48	114495.12	114495.26	28	1007.64	1008.14	1008.27	2309718.38	2309719.16	2309719.30
110151.42	110152.04	110152.17	29	1005.25	1005.73	1005.85	2195223.16	2195223.91	2195224.04
105937.27	105937.87	105938.00	30	1002.90	1003.35	1003.48	2085071.01	2085071.74	2085071.87
101848.24	101848.81	101848.94	31	1000.58	1001.01	1001.14	1979133.04	1979133.74	1979133.87
97880.65	97881.20	97881.33	32	998.28	998.70	998.82	1877284.12	1877284.80	1877284.93
94030.92	94031.47	94031.60	33	995.99	996.39	996.51	1779402.79	1779403.47	1779403.60
90295.63	90296.17	90296.30	34	993.69	994.08	994.21	1685371.20	1685371.87	1685372.00
86671.42	86671.97	86672.10	35	991.37	991.75	991.88	1595074.88	1595075.57	1595075.70
83155.10	83155.64	83155.78	36	989.01	989.40	989.53	1508402.77	1508403.46	1508403.60
79743.54	79744.09	79744.24	37	986.60	987.00	987.13	1425246.98	1425247.68	1425247.82
76433.74	76434.31	76434.45	38	984.13	984.53	984.68	1345502.73	1345503.44	1345503.58
73222.80	73223.38	73223.53	39	981.58	982.00	982.14	1269068.24	1269068.98	1269069.13
70107.90	70108.51	70108.66	40	978.92	979.36	979.51	1195844.69	1195845.45	1195845.60
67086.35	67087.00	67087.15	41	976.14	976.61	976.76	1125735.98	1125736.78	1125736.94
64155.55	64156.24	64156.39	42	973.22	973.72	973.88	1058648.78	1058649.63	1058649.79
61312.99	61313.72	61313.88	43	970.12	970.67	970.83	994492.34	994493.23	994493.39
58556.26	58557.05	58557.21	44	966.84	967.44	967.60	933178.40	933179.35	933179.51
55883.04	55883.90	55884.06	45	963.33	963.99	964.15	874621.12	874622.14	874622.30
53291.12	53292.05	53292.21	46	959.57	960.31	960.47	818736.99	818738.08	818738.24
50778.36	50779.38	50779.55	47	955.53	956.35	956.51	765444.68	765445.87	765446.03
48342.74	48343.85	48344.02	48	951.17	952.09	952.25	714665.04	714666.32	714666.48
45982.30	45983.53	45983.69	49	946.46	947.49	947.65	666320.90	666322.30	666322.46
43695.18	43696.55	43696.71	50	941.35	942.51	942.67	620337.08	620338.61	620338.77
41479.64	41481.15	41481.32	51	935.82	937.12	937.28	576640.21	576641.90	576642.06
39333.99	39335.68	39335.85	52	929.80	931.26	931.43	535158.71	535160.57	535160.74
37256.66	37258.54	37258.72	53	923.26	924.90	925.07	495822.66	495824.72	495824.90
35246.16	35248.25	35248.44	54	916.14	917.98	918.16	458563.70	458566.00	458566.18
33301.07	33303.42	33303.62	55	908.40	910.46	910.65	423314.97	423317.54	423317.74
31420.10	31422.73	31422.96	56	899.98	902.28	902.49	390011.01	390013.90	390014.12
29602.02	29604.98	29605.23	57	890.81	893.38	893.63	358587.67	358590.91	358591.16
27845.70	27849.02	27849.31	58	880.84	883.71	883.99	328981.98	328985.64	328985.94
26150.10	26153.82	26154.17	59	870.01	873.20	873.54	301132.14	301136.28	301136.62
24514.27	24518.44	24518.86	60	858.25	861.80	862.20	274977.36	274982.04	274982.45

Figure 10: Computational fuctions at 3%

22937.32	22942.00	22942.51	61	845.51	849.44	849.93	250457.80	250463.09	250463.60
21418.48	21423.74	21424.35	62	831.70	836.05	836.65	227514.47	227520.48	227521.09
19957.06	19962.94	19963.69	63	816.78	821.58	822.31	206089.19	206095.99	206096.74
18552.42	18559.00	18559.91	64	800.69	805.96	806.85	186124.45	186132.14	186133.05
17204.04	17211.37	17212.48	65	783.35	789.14	790.21	167563.36	167572.03	167573.14
15911.42	15919.59	15920.93	66	764.73	771.05	772.35	150349.54	150359.32	150360.66
14674.18	14683.25	14684.86	67	744.77	751.66	753.22	134427.12	134438.11	134439.72
13491.95	13502.00	13503.92	68	723.44	730.92	732.79	119740.62	119752.94	119754.86
12364.44	12375.54	12377.81	69	700.71	708.81	711.02	106234.90	106248.67	106250.94
11291.39	11303.60	11306.27	70	676.57	685.30	687.89	93855.13	93870.45	93873.13
10272.58	10285.94	10289.07	71	651.03	660.39	663.42	82546.75	82563.73	82566.86
9307.77	9322.35	9325.96	72	624.10	634.09	637.61	72255.45	72274.17	72277.79
8396.76	8412.57	8416.73	73	595.83	606.46	610.49	62927.13	62947.67	62951.83
7539.29	7556.36	7561.09	74	566.30	577.54	582.13	54507.96	54530.37	54535.10
6735.08	6753.40	6758.73	75	535.60	547.41	552.59	46944.39	46968.68	46974.01
5983.78	6003.31	6009.28	76	503.85	516.21	522.00	40183.16	40209.31	40215.28
5284.95	5305.64	5312.25	77	471.23	484.06	490.48	34171.44	34199.39	34206.00
4638.04	4659.78	4667.04	78	437.92	451.15	458.20	28856.86	28886.49	28893.75
4042.35	4065.02	4072.91	79	404.15	417.68	425.34	24187.67	24218.82	24226.71
3497.02	3520.46	3528.94	80	370.17	383.91	392.14	20112.86	20145.32	20153.80
3000.99	3024.99	3034.01	81	336.26	350.08	358.84	16582.36	16615.84	16624.86
2552.99	2577.32	2586.80	82	302.73	316.50	325.71	13547.18	13581.36	13590.84
2151.51	2175.90	2185.75	83	269.90	283.48	293.04	10959.70	10994.19	11004.04
1794.77	1818.94	1829.04	84	238.09	251.34	261.14	8773.81	8808.20	8818.30
1480.76	1504.40	1514.62	85	207.63	220.39	230.31	6945.21	6979.04	6989.26
1207.19	1230.00	1240.19	86	178.81	190.95	200.84	5431.63	5464.45	5474.64
971.53	993.21	1003.23	87	151.92	163.30	173.02	4193.08	4224.44	4234.45
771.01	791.31	800.98	88	127.19	137.70	147.10	3192.07	3221.55	3231.22
602.70	621.36	630.55	89	104.81	114.36	123.28	2393.81	2421.05	2430.24
463.48	480.33	488.91	90	84.89	93.42	101.74	1766.42	1791.12	1799.69
	365.09	372.93	91		74.96	82.57		1302.94	1310.78
		279.50	92			65.81			937.86
		205.54	93			51.45			658.36
		148.11	94			39.38			452.82
		104.41	95			29.47			304.71
		71.89	96			21.53			200.30
		48.27	97			15.32			128.41
		31.54	98			10.60			80.14
		20.02	99			7.12			48.60
		12.32	100			4.63			28.59
		7.33	101			2.90			16.27
		4.21	102			1.76			8.94
		2.33	103			1.02			4.73
		1.23	104			0.57			2.40
		0.63	105			0.30			1.17
		0.30	106			0.15			0.54
		0.14	107			0.07			0.24
		0.06	108			0.03			0.10
		0.03	109			0.01			0.04
		0.01	110			0.00			0.01

Age x	Duration 0 R[x]	Duration 1 R[x−1]+1	Duration 2+ Rx
17	58493.51		58495.04
18	57453.05	57454.22	57454.48
19	56416.10	56417.20	56417.45
20	55382.54	55383.57	55383.79
21	54352.21	54353.18	54353.38
22	53325.02	53325.91	53326.10
23	52300.81	52301.65	52301.82
24	51279.47	51280.27	51280.43
25	50260.93	50261.67	50261.83
26	49245.07	49245.78	49245.92
27	48231.82	48232.49	48232.63
28	47221.12	47221.74	47221.88
29	46212.88	46213.48	46213.61
30	45207.04	45207.62	45207.75
31	44203.58	44204.14	44204.27
32	43202.47	43203.00	43203.13
33	42203.65	42204.18	42204.31
34	41207.14	41207.66	41207.79
35	40212.93	40213.45	40213.58
36	39221.03	39221.56	39221.70
37	38231.49	38232.02	38232.16
38	37244.34	37244.89	37245.03
39	36259.64	36260.21	36260.35
40	35277.47	35278.06	35278.21
41	34297.92	34298.55	34298.70
42	33321.11	33321.78	33321.93
43	32347.19	32347.90	32348.05
44	31376.30	31377.06	31377.22
45	30408.64	30409.46	30409.62
46	29444.41	29445.31	29445.47
47	28483.85	28484.84	28485.00
48	27527.25	27528.33	27528.49
49	26574.89	26576.08	26576.24
50	25627.11	25628.43	25628.59
51	24684.29	24685.76	24685.92
52	23746.84	23748.47	23748.64
53	22815.22	22817.04	22817.21
54	21889.93	21891.96	21892.14
55	20971.50	20973.78	20973.98

Figure 11: Computational function at 3%

56	20060.55	20063.10	20063.32
57	19157.72	19160.58	19160.83
58	18263.70	18266.91	18267.20
59	17379.26	17382.86	17383.20
60	16505.21	16509.25	16509.66
61	15642.43	15646.96	15647.45
62	14791.84	14796.92	14797.52
63	13954.46	13960.14	13960.87
64	13131.32	13137.67	13138.56
65	12323.55	12330.63	12331.71
66	11532.31	11540.19	11541.50
67	10758.82	10767.58	10769.14
68	10004.35	10014.05	10015.91
69	9270.22	9280.91	9283.12
70	8557.74	8569.50	8572.10
71	7868.30	7881.17	7884.21
72	7203.24	7217.27	7220.79
73	6563.92	6579.14	6583.18
74	5951.67	5968.09	5972.69
75	5367.76	5385.37	5390.56
76	4813.39	4832.16	4837.96
77	4289.66	4309.54	4315.96
78	3797.54	3818.43	3825.48
79	3337.85	3359.62	3367.28
80	2911.21	2933.70	2941.94
81	2518.01	2541.03	2549.79
82	2158.41	2181.74	2190.95
83	1832.29	1855.67	1865.24
84	1539.22	1562.38	1572.19
85	1278.47	1301.12	1311.05
86	1048.98	1070.84	1080.74
87	849.39	870.17	879.89
88	678.04	697.47	706.87
89	532.97	550.85	559.77
90	412.03	428.16	436.49
91		327.14	334.75
92			252.18
93			186.37
94			134.92
95			95.53
96			66.06
97			44.53
98			29.21
99			18.60
100			11.48
101			6.86
102			3.95
103			2.19
104			1.16
105			0.59
106			0.29
107			0.13
108			0.06
109			0.02
110			0.01